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DEPARTAMENTO DE ELECTRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA
LICENCIATURA EM ENG. DE COMPUTADORES E INFORMÁTICA

REDES DE COMUNICAÇÕES I

GUIA PRÁTICO 1 – EXPERIÊNCIAS BASE

Objectives

- Verify the network configuration of a PC
- Name translation to IP addresses and vice-versa
- Connectivity tests
- Discovery of the path between two network machines
- Geo-location and discovery of the entity and responsible for the network machines

Duration

1 class

1. Verify the network configuration of a PC

1.1. Open a command window and execute the command: *ipconfig* and register:
(i) how many network interfaces (Linux: *ip link;* MAC: *ifconfig*) and
(ii) the IP address and the *default gateway* of each interface (Linux: *ip addr* and *ip route;* MAC: *netstat -r*).

1.2. Repeat the command *ipconfi*g with the option */all* ("ipconfig /all")and register:
(i) the name of the machine (Linux: *cat /etc/hostname*)
(ii) the physical address of each interface (Linux: *ip addr*) and
(iii) the DNS servers IP addresses (Linux: *cat /etc/resolv.conf*).

2. Name translation to IP addresses and vice-versa

2.1. In a command window, using the command *nslookup* (same in Linux), determine the IP addresses associated to each of the following names:

Name	IP Address(es)
www.ua.pt	193.136.173.58
ua.pt	193.126.172.175 193.136.172.174 193.136.172.173
www.tvi.iol.pt	193.126.240.138
www.sapo.pt	213.13.146.142
www.tsf.pt	2606:4700:20::681a:7ed 2606:4700:20::681a:6ed 2606:4700:20::ac43:4942 10
www.antena3.pt	94.46.160.176
www.rtp.pt	146.75.90.192
www.publico.pt	108.157.98.97 108.157.98.54 108.157.98.35 108.157.98.113
www.google.com	2a00:1450:4003:80d::2004 142.250.200.68
www.google.pt	2a00:1450:4003:806::2003 216.58.215.163
www.google.es	2a00:1450:4003:80e::2003 142.250.200.99
www.google.fr	2a00:1450:4003:800::2003 216.58.215.131

```
nslookup [{address_to_find}]
```

2.2. Using the command *nslookup* determine the name associated to the following IP addresses:

IP Address	Name
193.136.173.58	lvs-ng.ua.pt
193.137.55.13	www.up.pt
157.240.212.35	edge-star-mini-shv-01-lis1.facebook.com
31.13.66.174	instagram-p42-shv-01-iad3.fbcdn.net

2.3. Open the browser and access to each of the following URLs:

Addresses
193.137.55.13
157.240.212.35

3. Connectivity tests

3.1. In a command window execute the command *ping* (same in Linux) to the following addresses, and register the average round trip time. What can you conclude about the relation between the round trip time and the geographical distance? Note: If pings do not work, connect through the WiFi network in the lab.

Addresses	Machine location	Average round trip time
www.ua.pt	Aveiro, Portugal (0Km)	7ms
www.up.pt	Porto, Portugal (~60Km)	11ms
www.fc.ul.pt	Lisboa, Portugal (~220Km)	timed out
www.utad.pt	Vila Real, Portugal (~160Km)	timed out
www.uevora.pt	Évora, Portugal (~250Km)	timed out
www.uam.es	Madrid, Espanha (~420Km)	29ms
www.univ-paris8.fr	Paris, França (~1260Km)	timed out
web.mit.edu	EUA (~5100Km)	57ms
www.unisa.ac.za	África do Sul (~8750Km)	timed out
www.adelaide.edu.au	Austrália (~17100Km)	359ms
www.tanzaniatourism.go.tz	Tanzânia (~3100Km)	timed out

4. Discovery of the path between two network machines

4.1. In a command window execute the command *tracert* (Linux: *traceroute*) to the following addresses, and register the number of network machines between the origin and destination, and the address of the antepenultimate machine in the path. Repeat using the option *–d* of the *tracert* command.

Addresses	Machine location	Number of machines	IP address of the antepenultimate machine in the path
www.ua.pt	Aveiro, Portugal (0Km)	4	192.168.63.254
www.up.pt	Porto, Portugal (~60Km)	11	194.210.6.205
www.fc.ul.pt	Lisboa, Portugal (~220Km)	12	193.137.1.18
www.utad.pt	Vila Real, Portugal (~160Km)	13	193.137.3.135
www.uevora.pt	Évora, Portugal (~250Km)	11	194.210.6.207
www.uam.es	Madrid, Espanha (~420Km)	16	193.145.14.30
www.univ-paris8.fr	Paris, França (~1260Km)	16	193.51.177.109
web.mit.edu	EUA (~5100Km)	15	80.255.14.1
www.unisa.ac.za	África do Sul (~8750Km)	20	155.232.1.153
www.adelaide.edu.au	Austrália (~17100Km)	18	113.197.15.44
jornalnopintcha.gw	Guiné Bissau (~3100Km)	25	213.248.85.37

5. Discovery of the entity and responsible for the network machines

5.1. Using the service *whois*, through the web page https://who.is/ (or https://ping.eu/ns-whois/), determine (if possible), for each of the *trace routes* in 4.1: the entity responsible by the **antepenultimate** machine of each path and the location of that entity.

IP address	Responsible entity	Location of the entity
192.168.63.254	Internet Assigned Numbers Authority	Los Angeles
194.210.6.205	RIPE Network Coordination Centre	Amsterdam
193.137.1.18	RIPE Network Coordination Centre	Amsterdam
193.137.3.135	RIPE Network Coordination Centre	Amsterdam
194.210.6.207	RIPE Network Coordination Centre	Amsterdam
193.145.14.30	RIPE Network Coordination Centre	Amsterdam
193.51.177.109	RIPE Network Coordination Centre	Amsterdam
80.255.14.1	RIPE Network Coordination Centre	Amsterdam
155.232.1.153	African Network Information Center	Ebene
113.197.15.44	Asia Pacific Network Information Centre	South Brisbane
213.248.85.37	RIPE Network Coordination Centre	Amsterdam

6. Geo-location of IP addresses

6.1. Using the geo-location service, through the web page http://www.hostip.info, determine for each of the *trace routes* in 5, the geographic location of the **antepenultimate** machine of each path. Note: this service is not precise.

Addresses	Location	
	Country	City
192.168.63.254		
194.210.6.205	PORTUGAL	
193.137.1.18	PORTUGAL	Lisboa
193.137.3.135	PORTUGAL	
194.210.6.207	PORTUGAL	
193.145.14.30	SPAIN	Huddinge
193.51.177.109	FRANCE	
80.255.14.1	FRANCE	
155.232.1.153	SOUTH AFRICA	
113.197.15.44		
213.248.85.37	EUROPEAN UNION	